Rejection Sampling is Optimal for Relative Entropy Coding

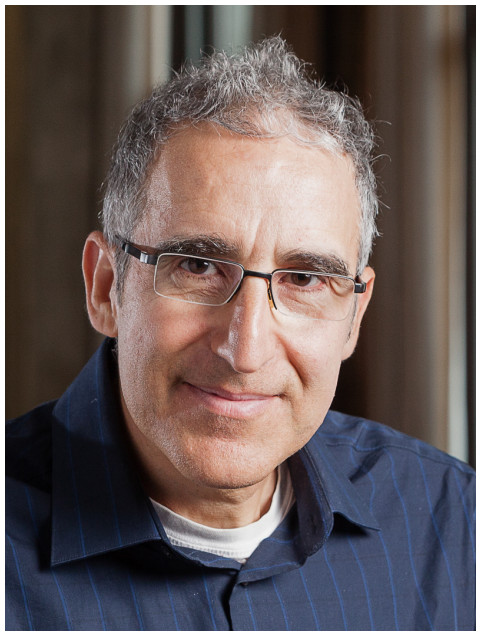
Spencer Hill

Queen's University, Canada

Information Theory Workshop

November 11, 2026

In collaboration with



Fady Alajaji

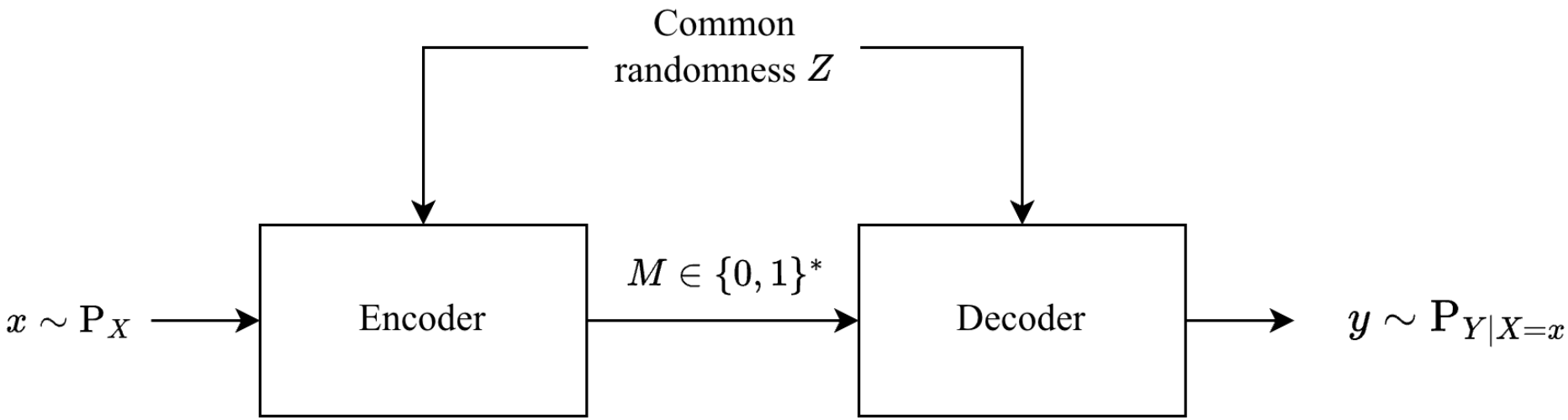


Tamás Linder



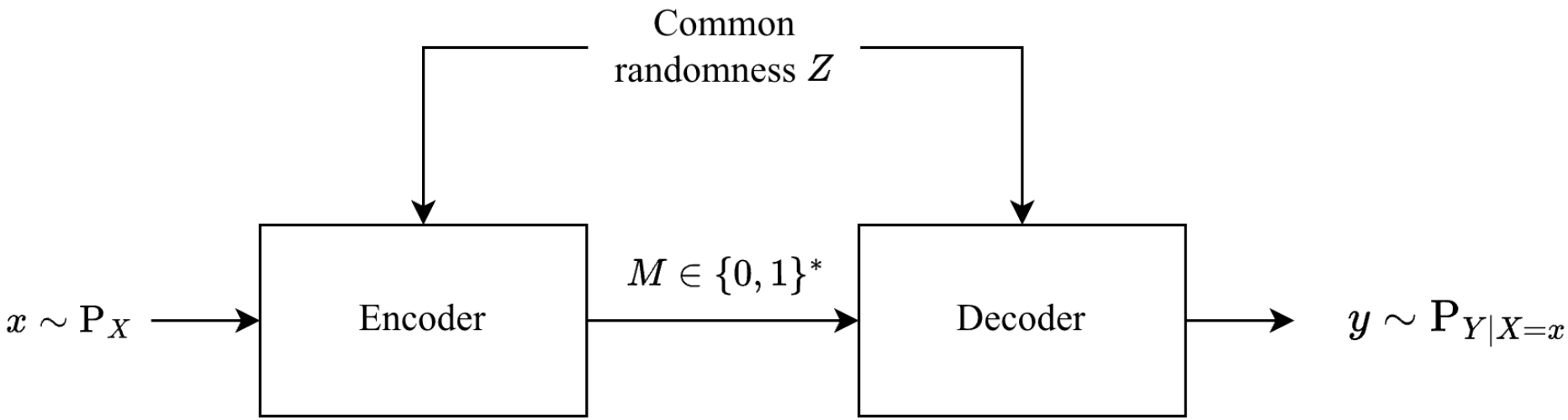
Gergely Flamich

Relative Entropy Coding (Channel Simulation)



Ingredients:

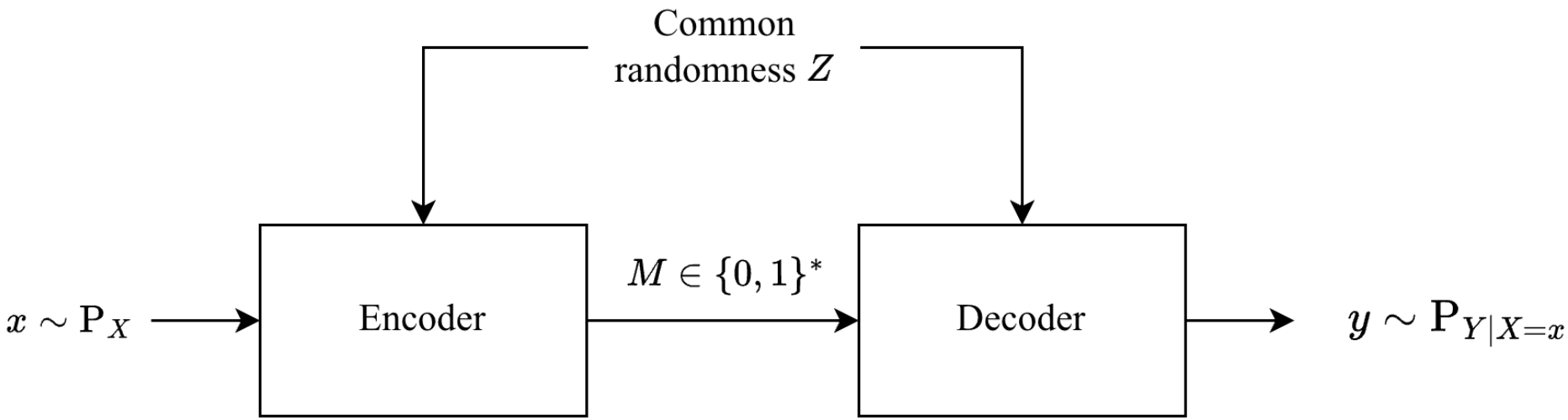
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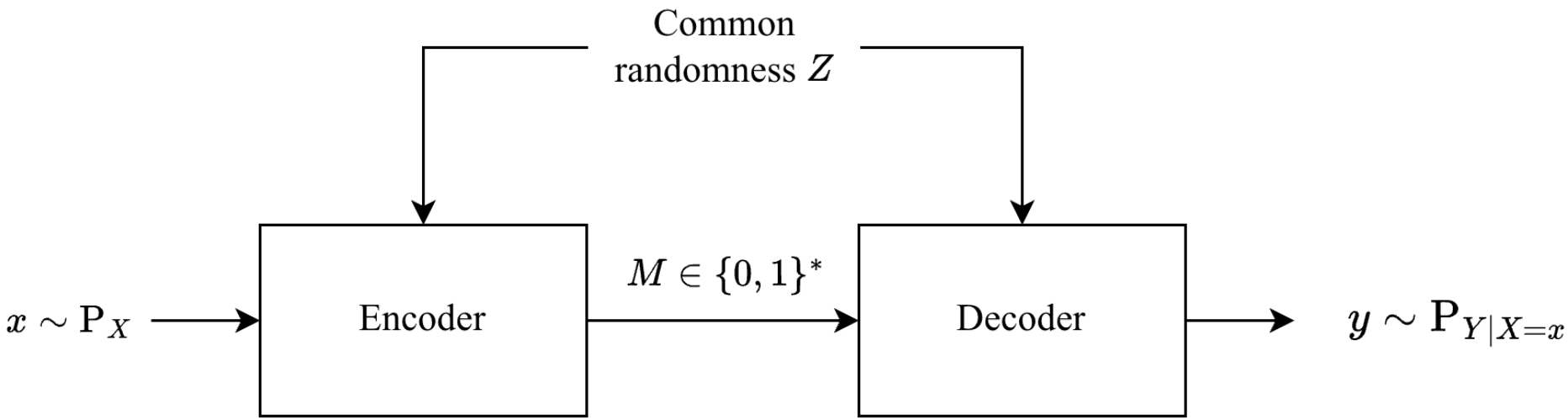
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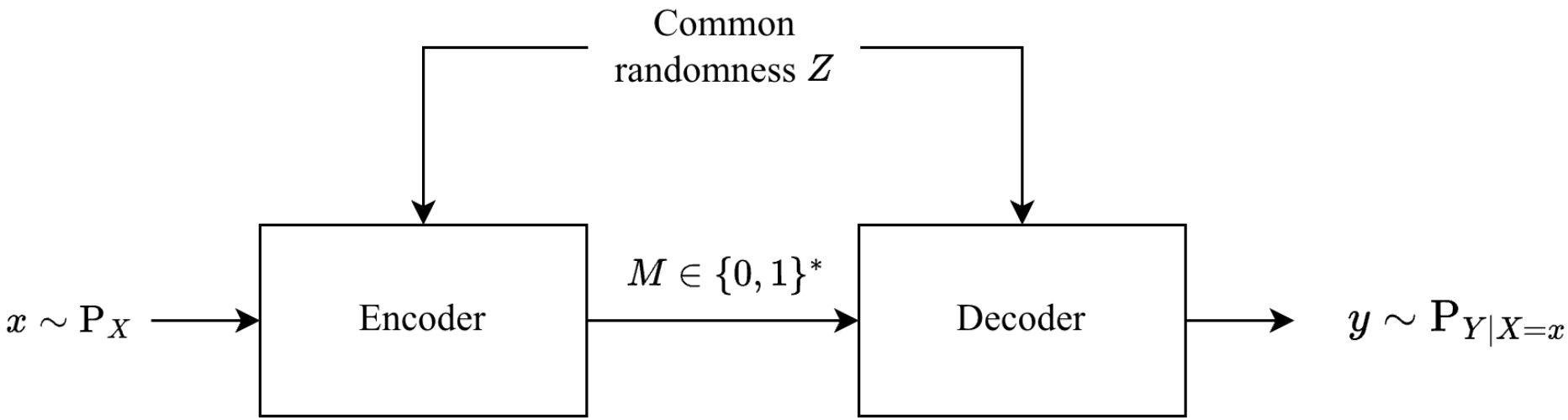
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- **Exact Requirement:** $\mathrm{dec}(\mathrm{enc}(x, Z), Z) \sim P_{Y|X=x}$ for every x

Objective and Bounds

Objective:

$$R^* := \min_{(Z,\text{enc},\text{dec})} \mathbb{E}[|\text{enc}(X,Z)|] \quad \text{s.t.} \quad \text{dec}(\text{enc}(x,Z),Z) \sim P_{Y|X=x}.$$

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This work: For $I_{\text{F}}(X \to Y)$ the *functional information*,

$$I_{\text{F}}(X \to Y) \leq R^* \leq I_{\text{F}}(X \to Y) + 2.45,$$

and the upper bound is achieved by *rejection sampling*.

Outline

- The functional information
- Rejection sampling and the ring toss code
- Application: tight one-shot lossy source coding

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Definition (Width Function)

For two distributions $P \ll Q$, the Q-width function at height level $\ell \geq 0$ is defined as

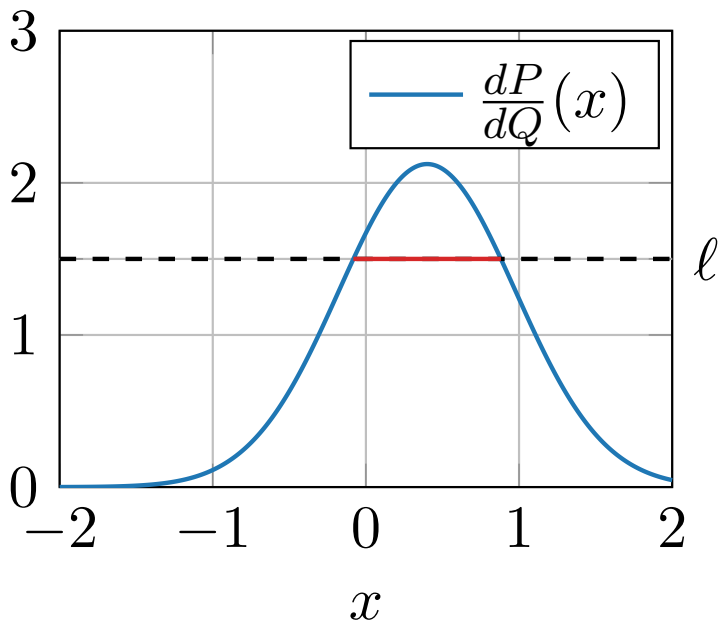
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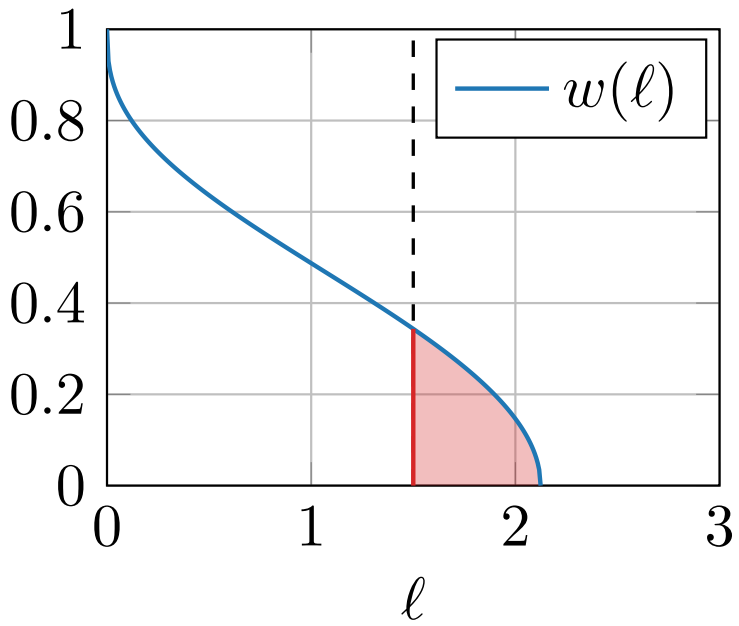
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Define the y-width function ($Q = P_X$ and $P = P_{X|Y}(\cdot \mid y)$)

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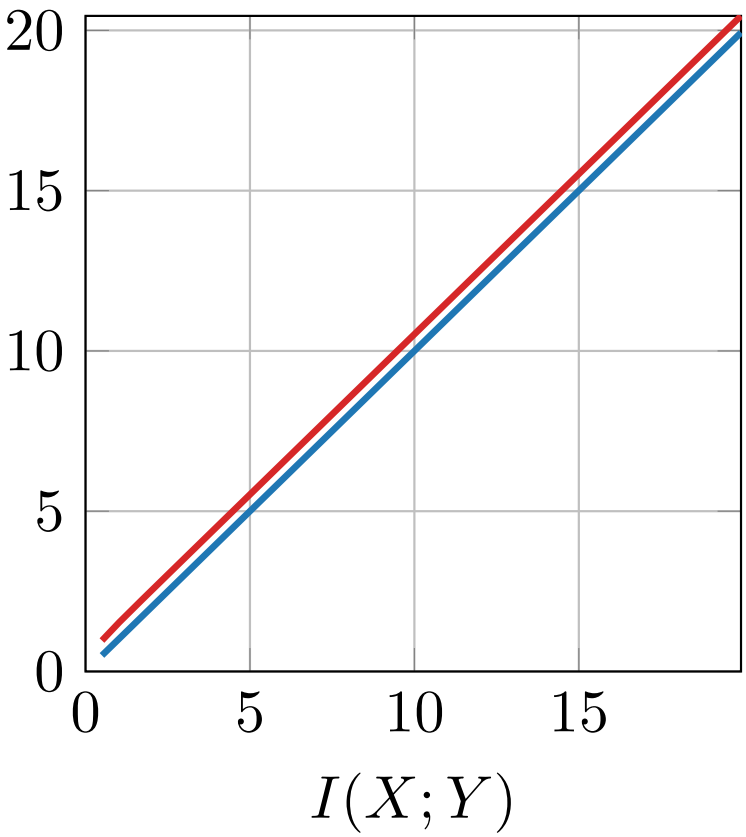
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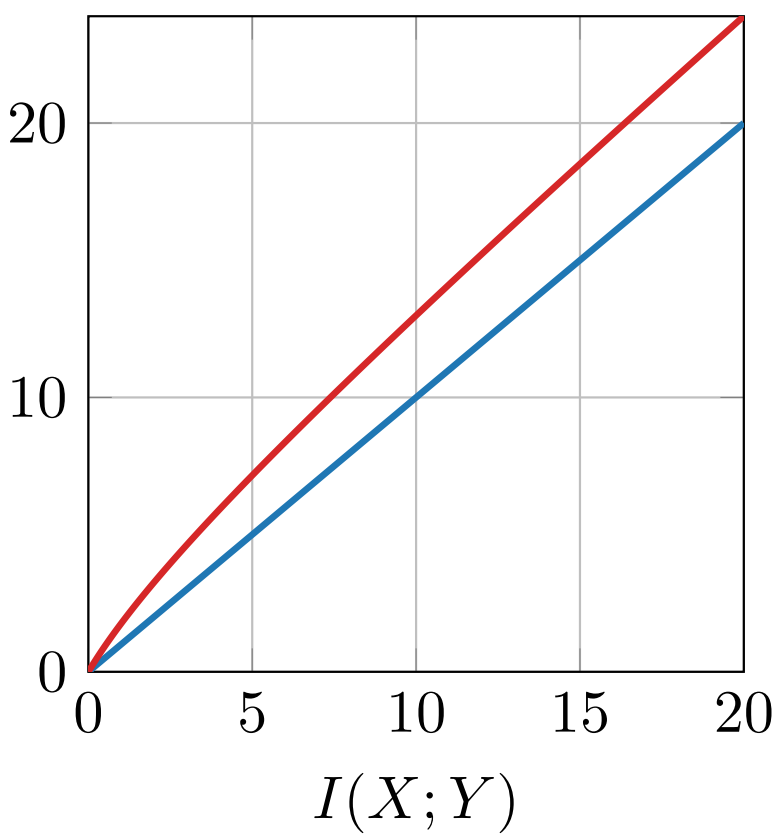
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- $I_{\mathrm{F}}(X \to Y) \leq R^*$ (Flamich, Sriramu, and Wagner 2025)

Some Examples

AWGN



Particular Additive Noise mod 1



$I(X;Y)$ $I_{\mathrm{F}}(X \to Y)$

Outline

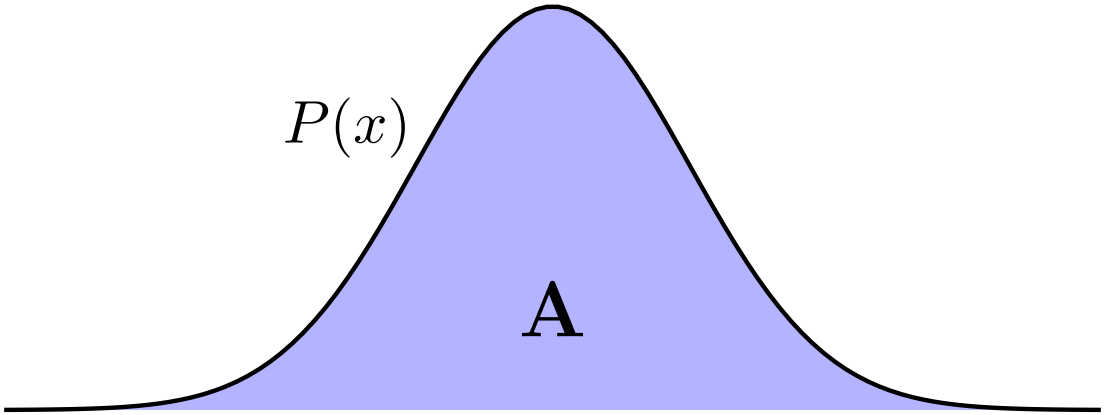
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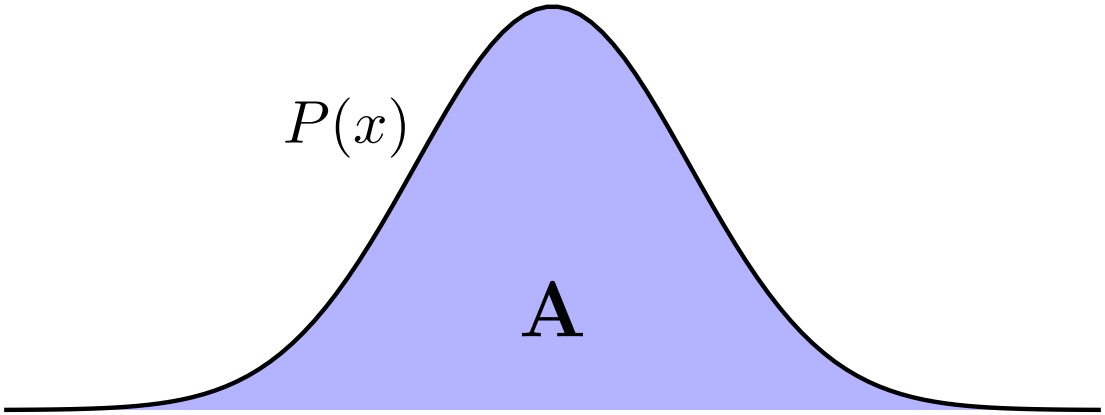
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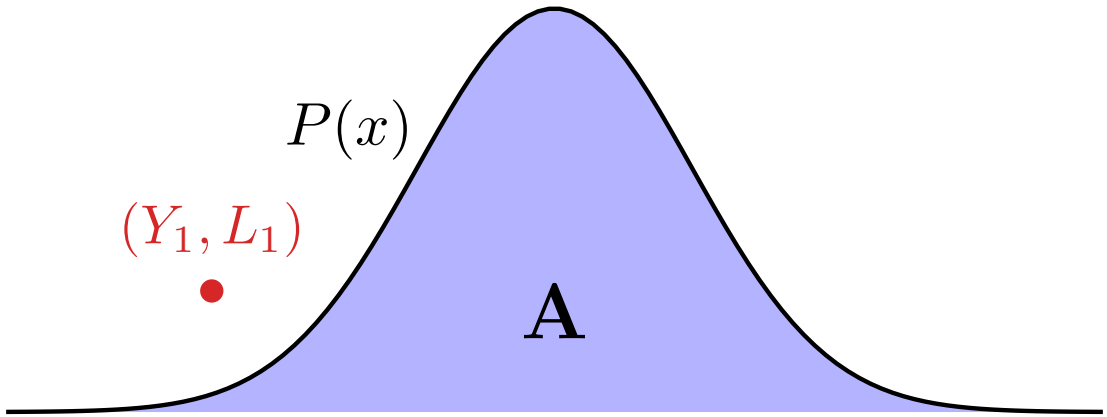
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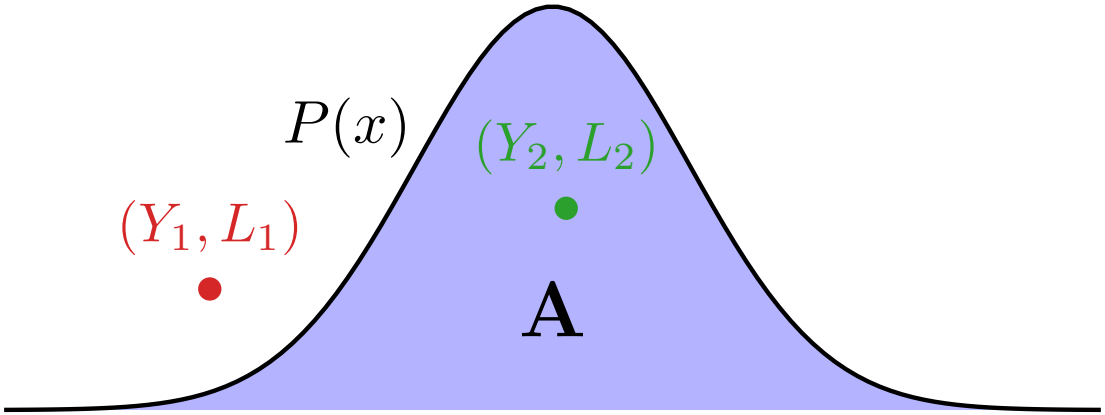
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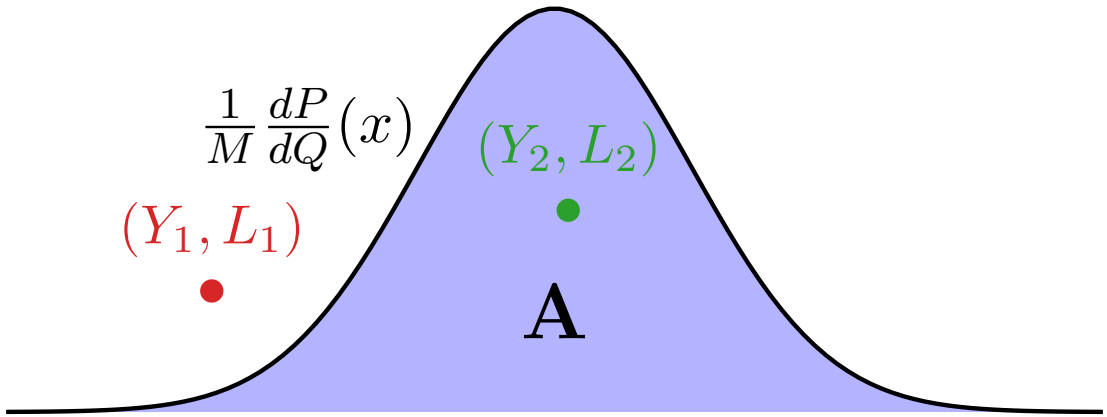
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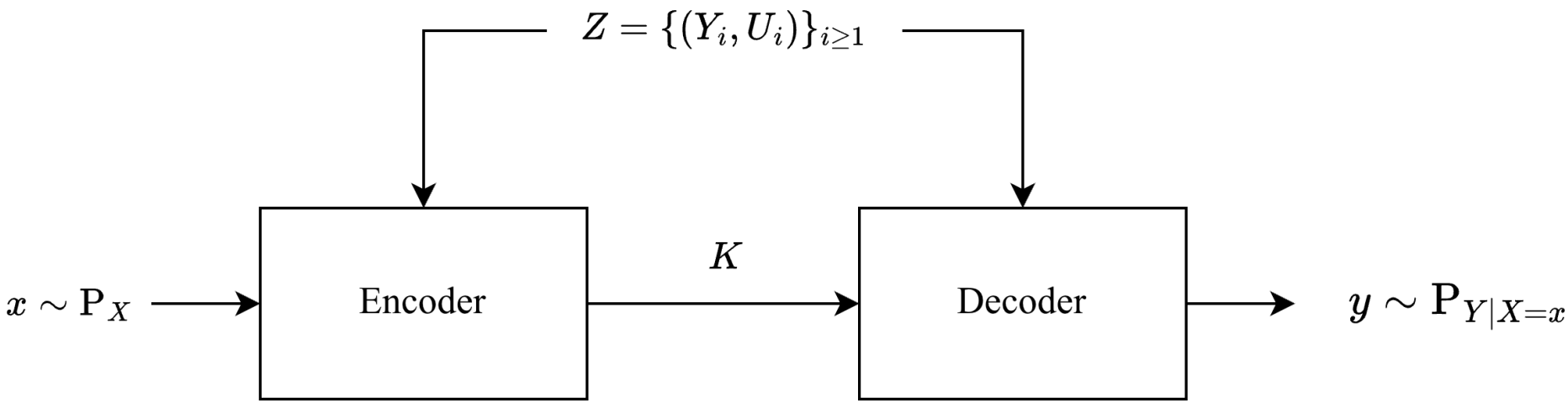
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- For general Q: accept Y_i if

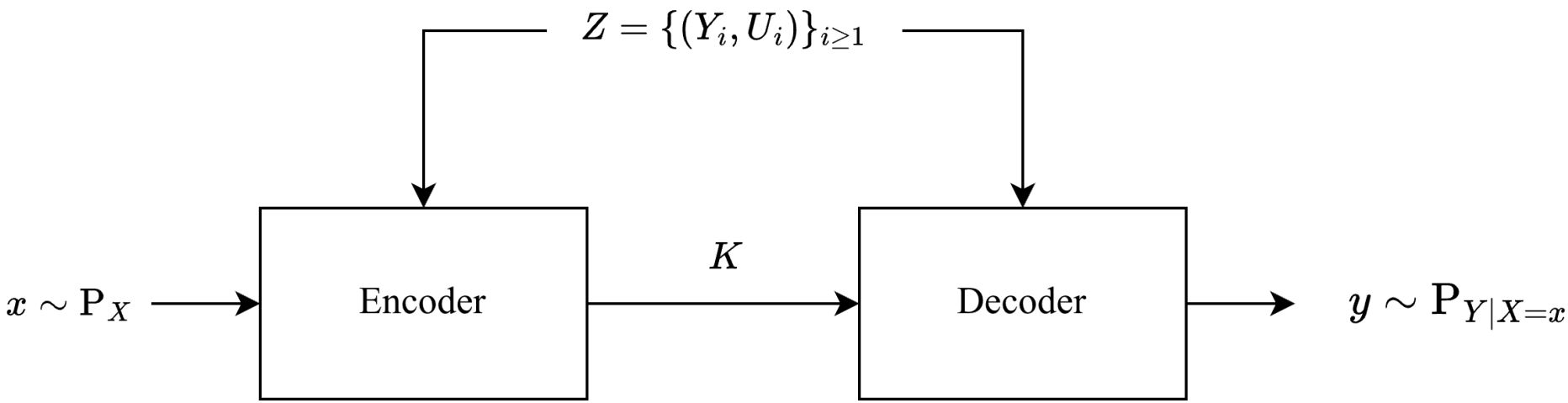
$$U_i \leq \frac{1}{M}\frac{dP}{dQ}(Y_i), \qquad \left\|\frac{dP}{dQ}\right\|_\infty \leq M.$$

Rejection Sampling as a Relative Entropy Code



On input x:

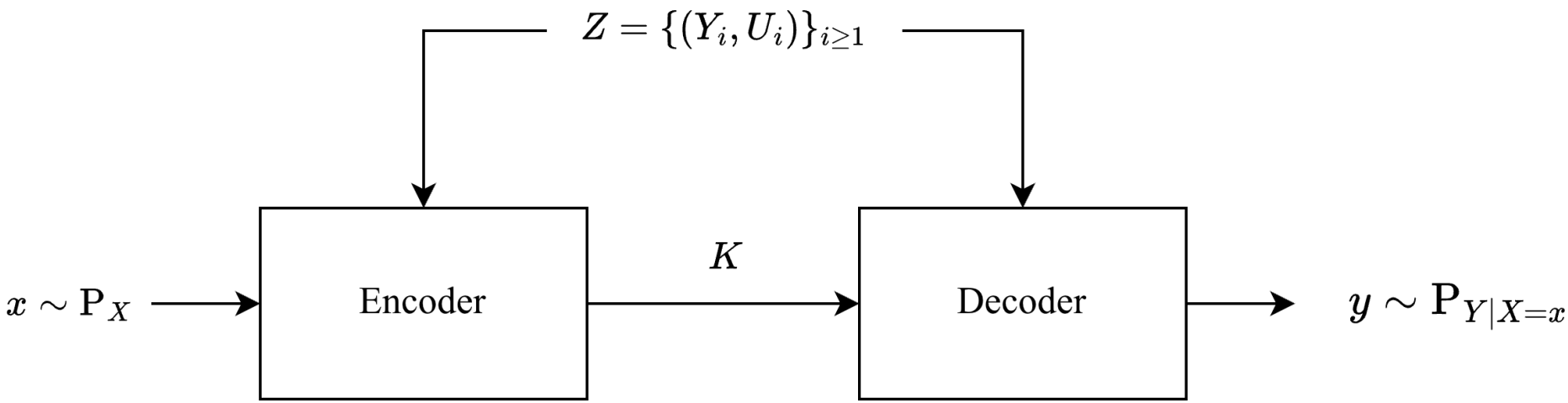
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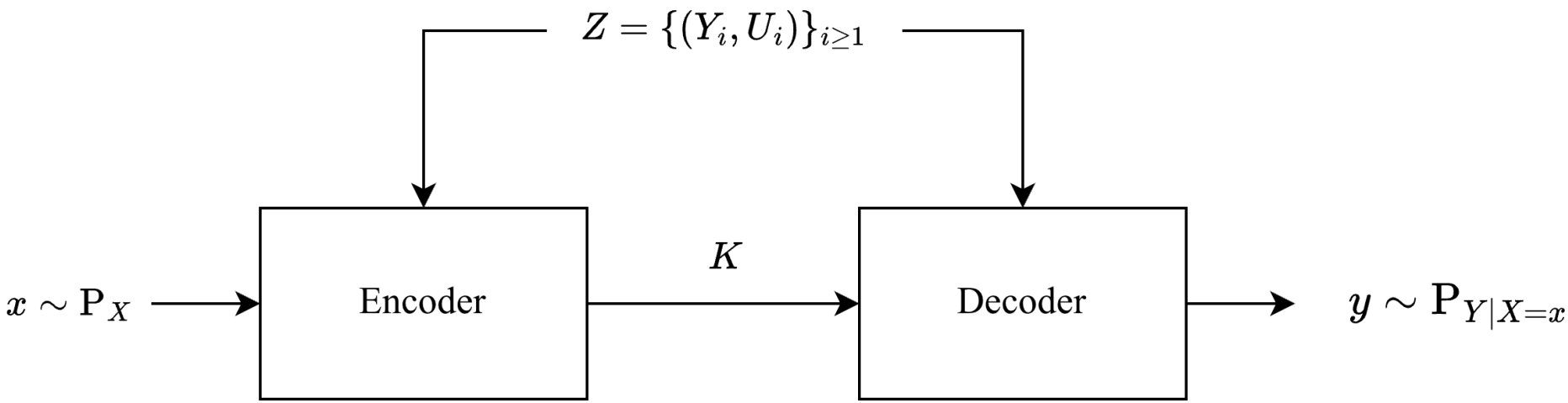
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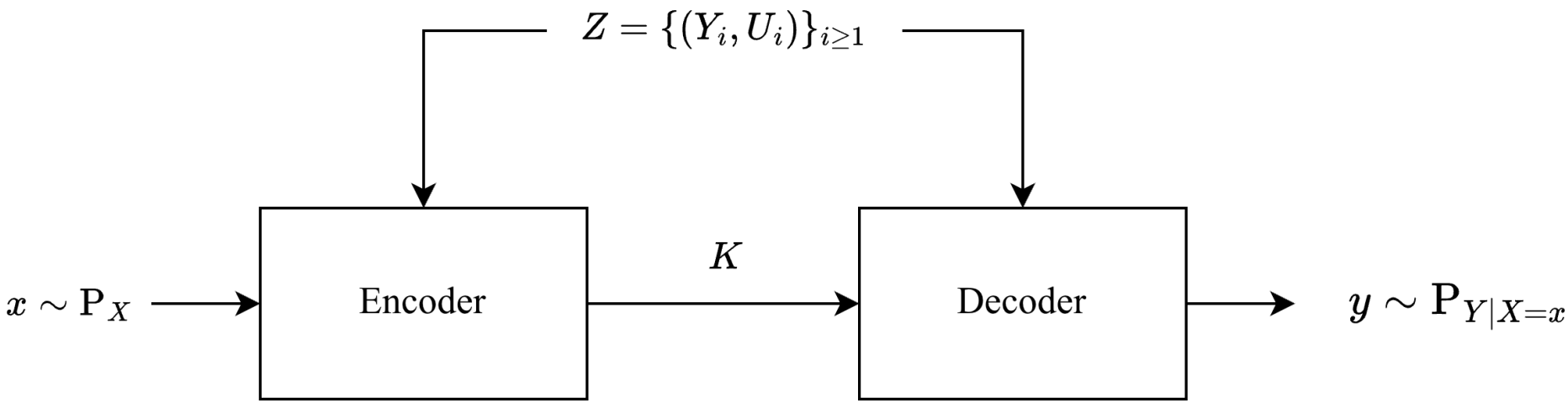
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- Send $K = \min \left\{ i : U_i \leq \frac{1}{M} \frac{dP_{Y|X}}{dP_Y}(Y_i \mid x) \right\}$

Encoding the Index K

- Unconditionally, $K \sim \text{Geom}(1/M)$

Encoding the Index K

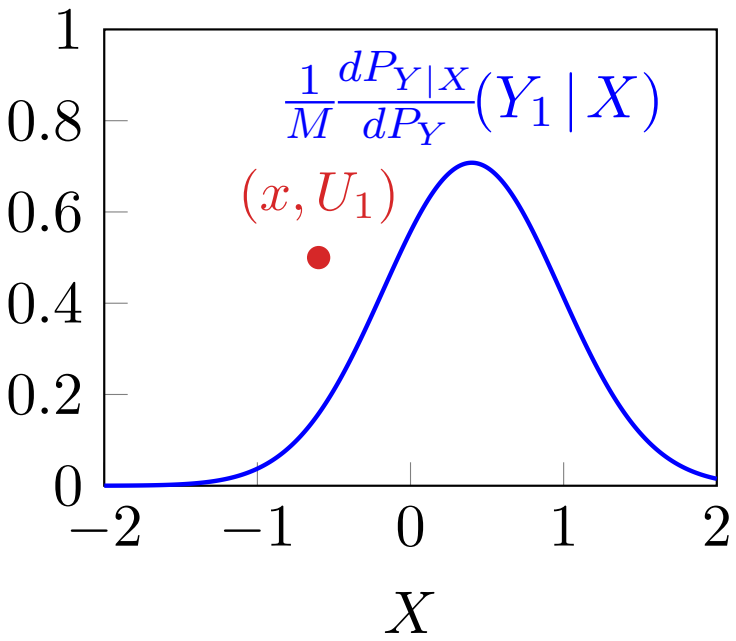
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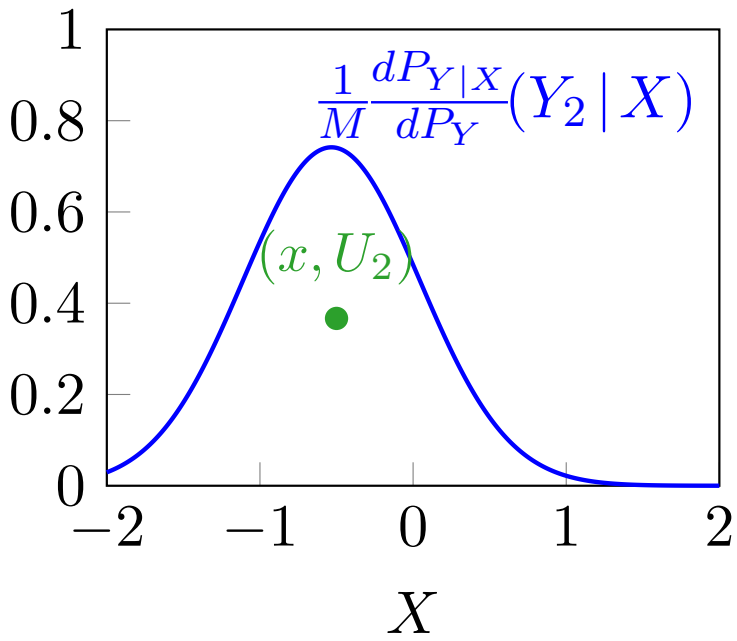
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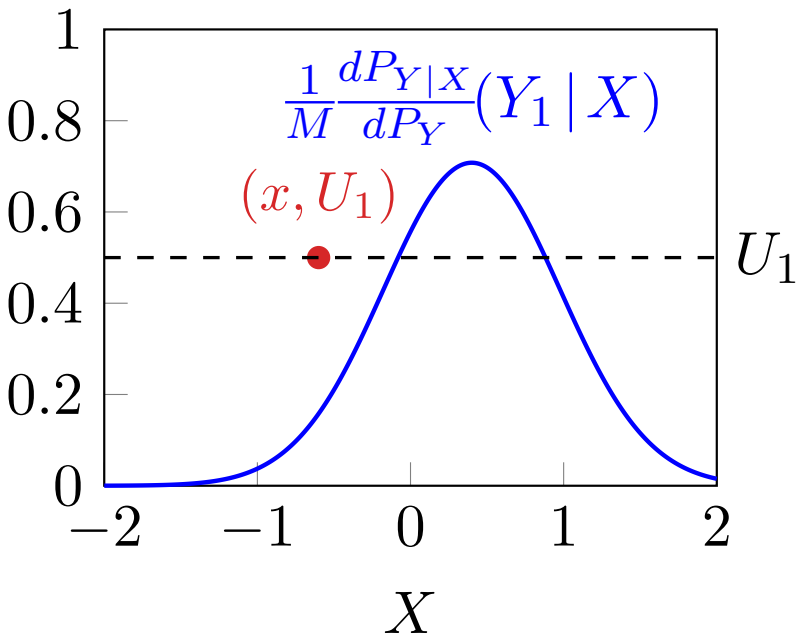
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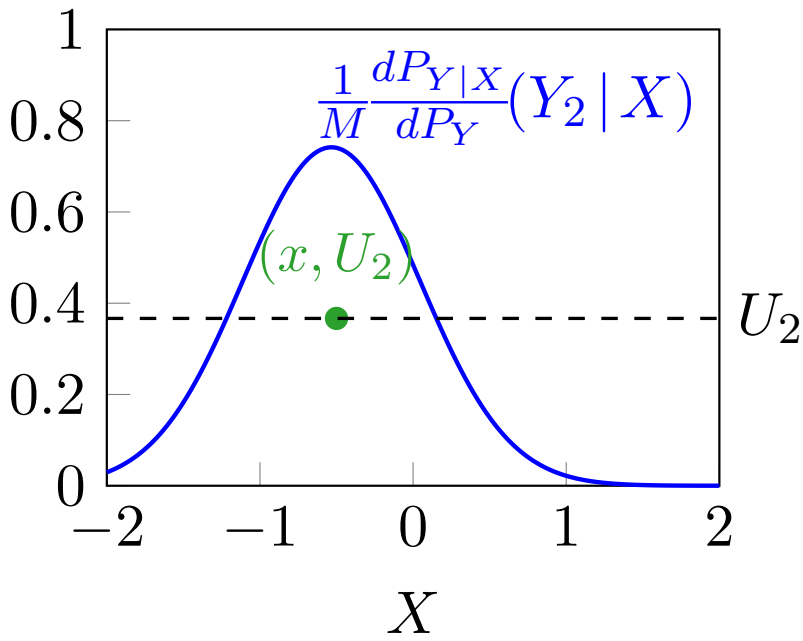




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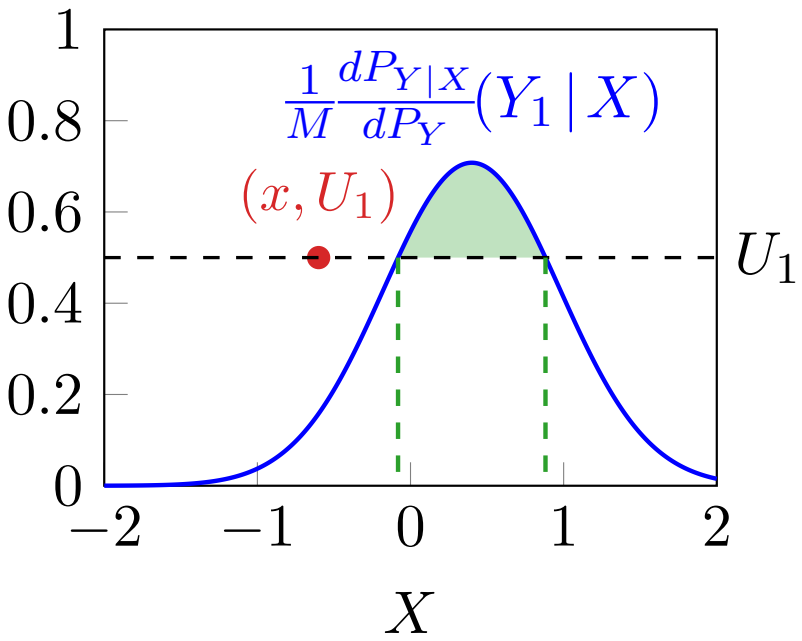
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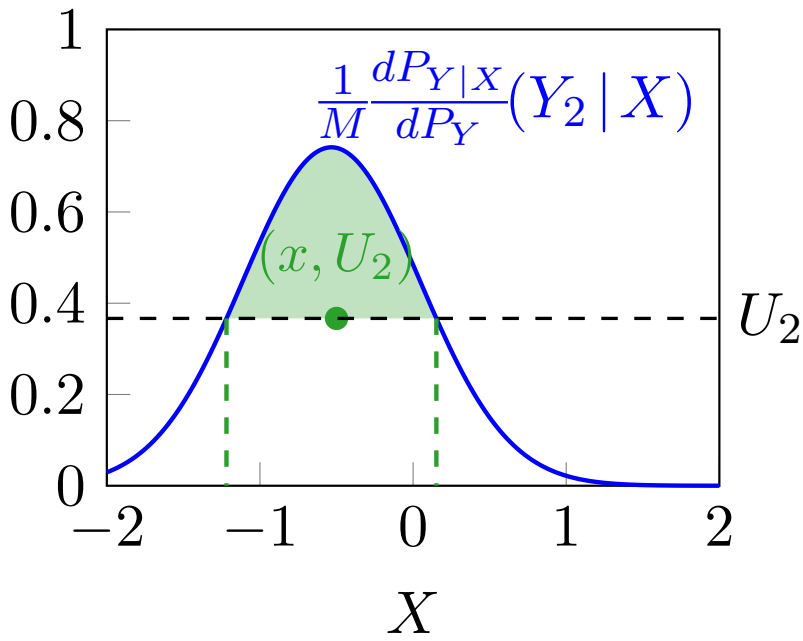




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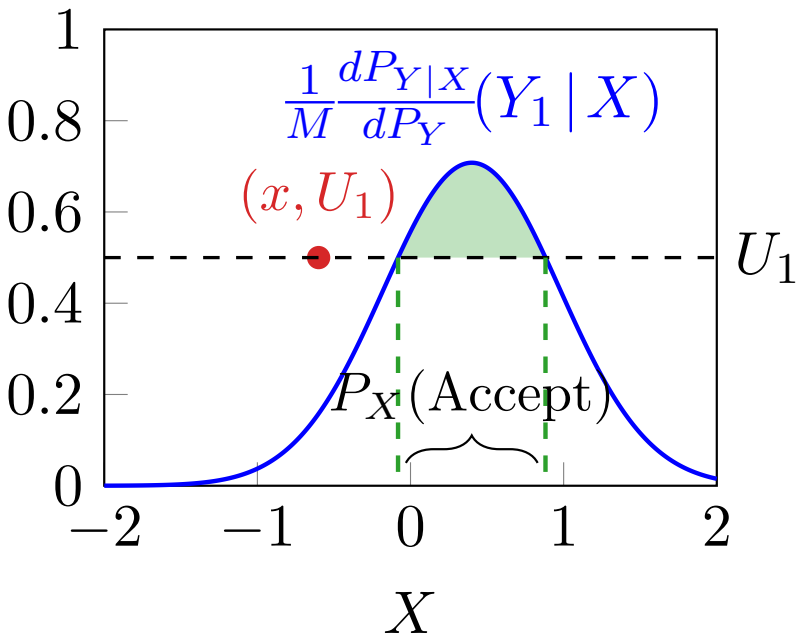
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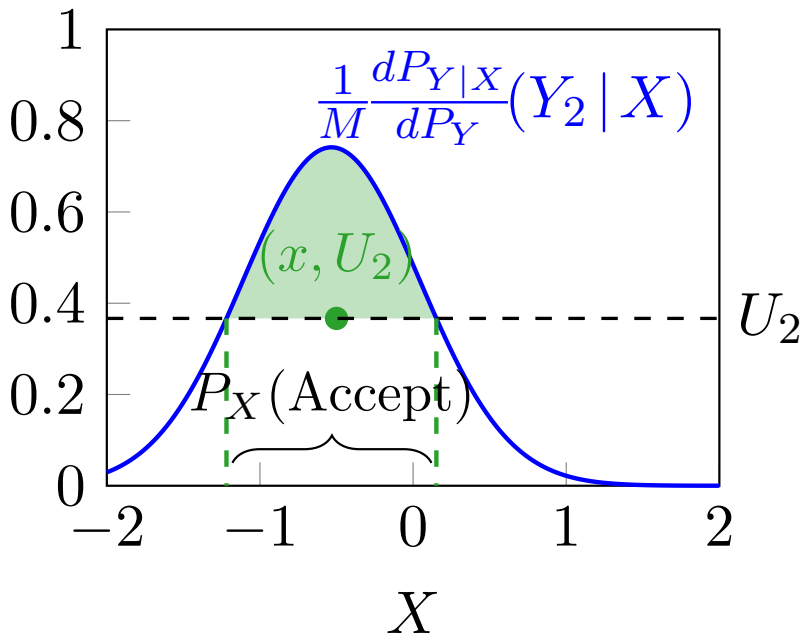




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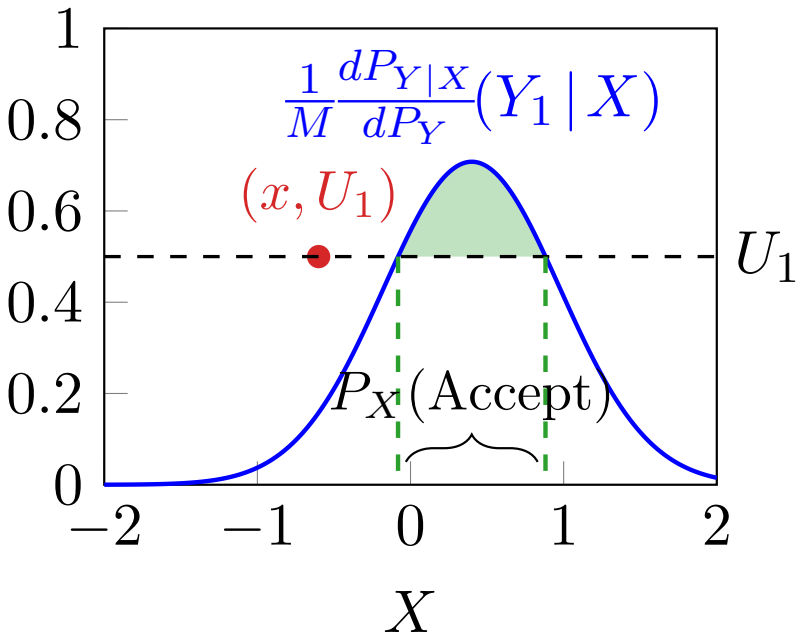
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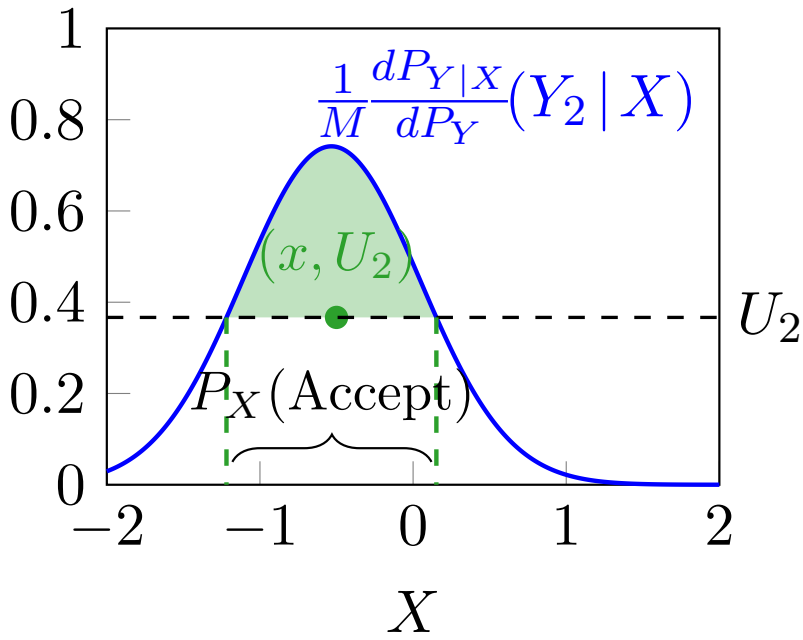




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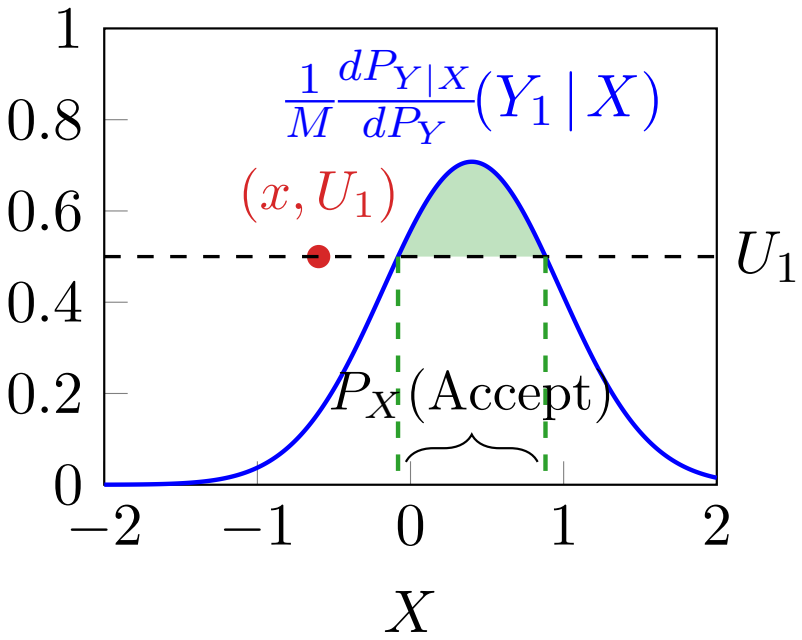


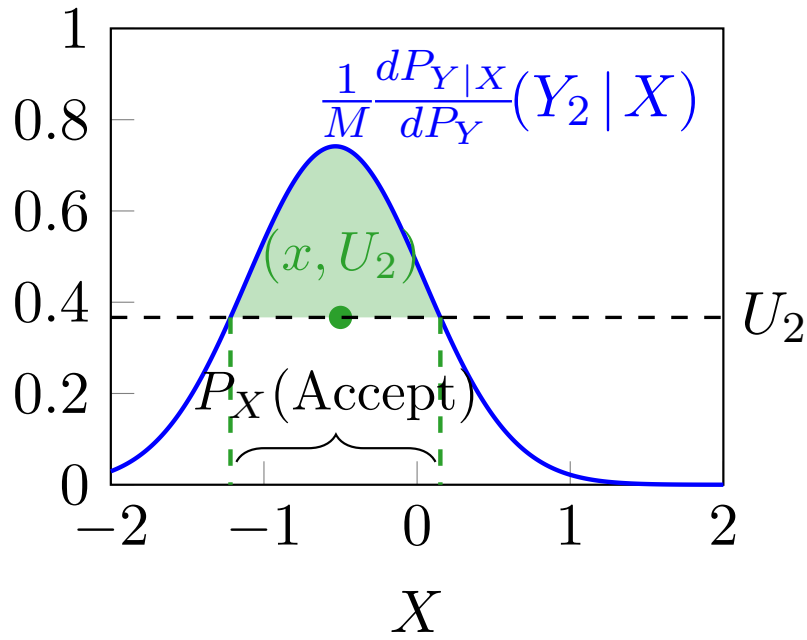


$P_X(\text{Accept } i)$

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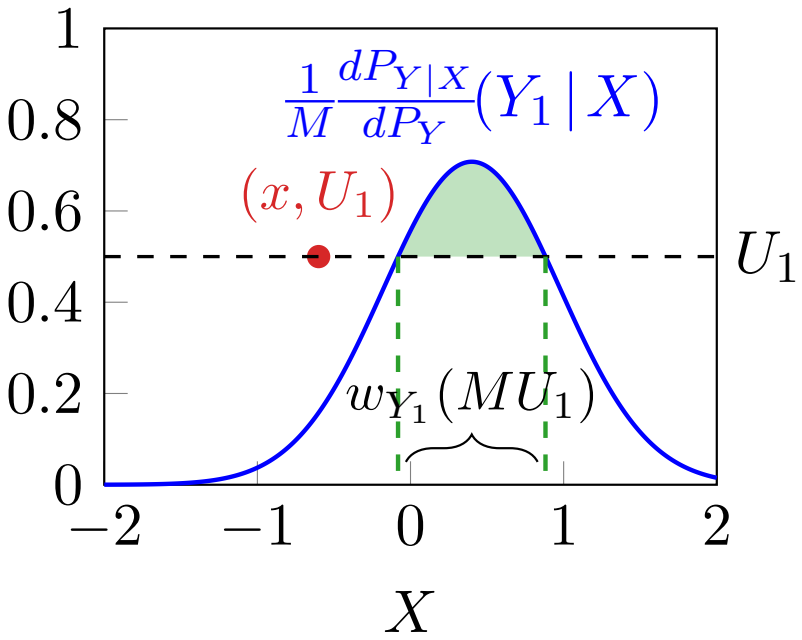


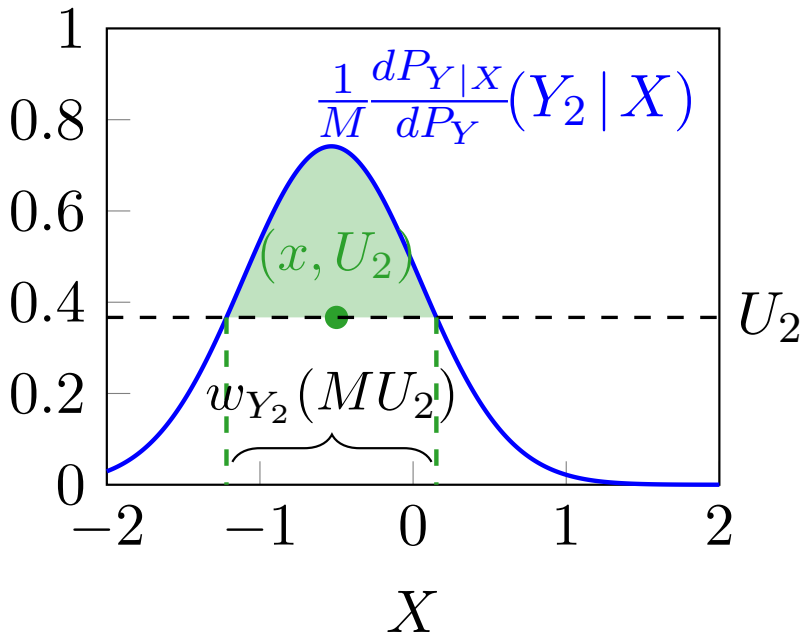


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Encode K using a Shannon-Fano code optimal for the distribution

$$Q_{K|Z}(k \mid z) = w_{y_k}(Mu_k) \prod_{i=1}^{k-1} (1 - w_{y_i}(Mu_i)).$$

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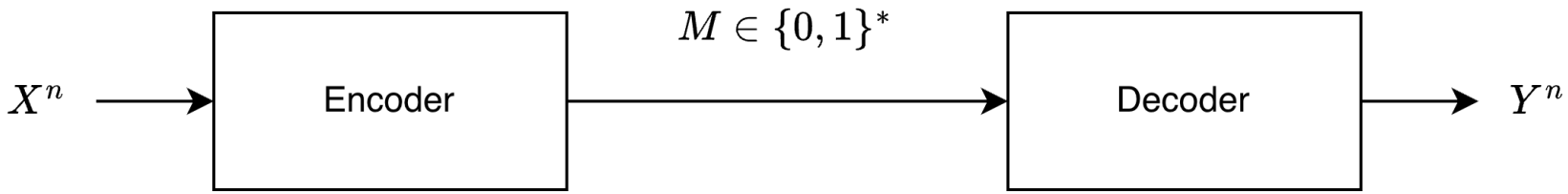
- **Tight characterization:** $I_{\mathrm{F}}(X \to Y) \leq R^* \leq I_{\mathrm{F}}(X \to Y) + 2.45$
- **Improved mutual information bound:**

$$I(X;Y) \leq R^* \leq I(X;Y) + \log(I(X;Y) + 1) + 3.45.$$

Outline

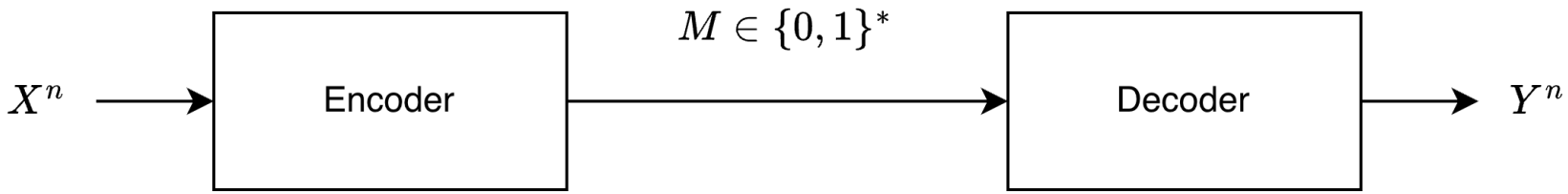
- The functional information
- Rejection sampling and the ring toss code
- Application: tight one-shot lossy source coding

Lossy Source Coding



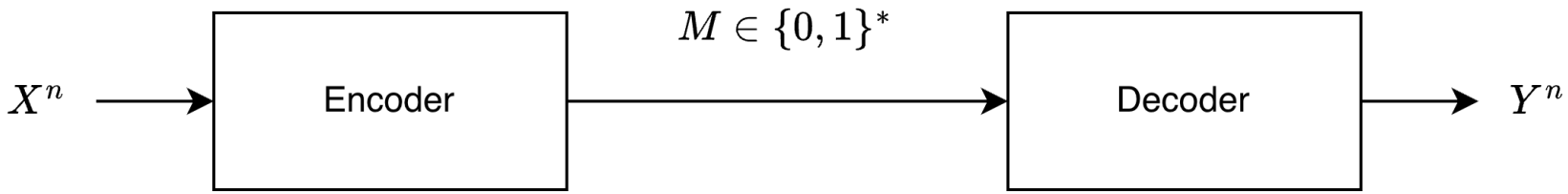
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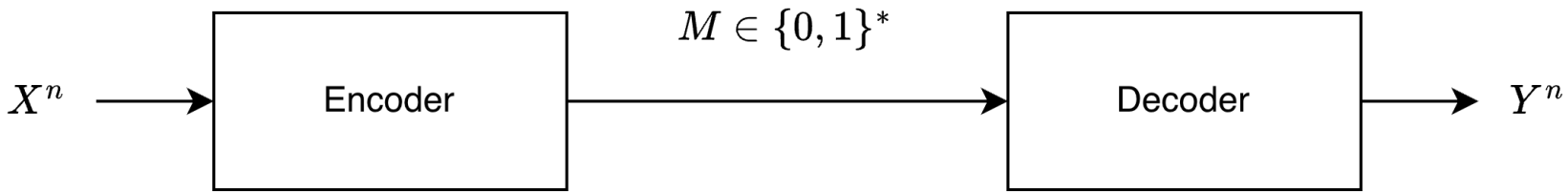
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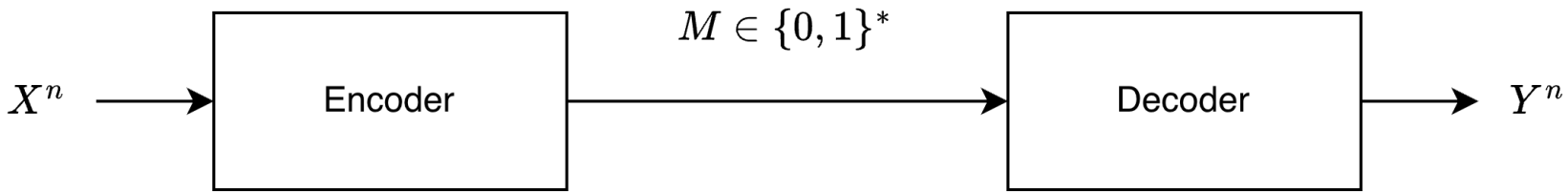
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- Shannon (1948): *Asymptotically* ($n \to \infty$) optimal performance:

$$R(D) = \min_{\mathrm{P}_{Y|X} \,:\, \mathbb{E}[\Delta(X,Y)] \leq D} I(X;Y).$$

Optimal One-Shot Lossy Source Coding

One-shot rate distortion function:

$$R_O(D) := \inf_{(\text{enc},\text{dec})} \mathbb{E}[|\text{enc}(X)|] \quad \text{s.t.} \quad \mathbb{E}[\Delta(X,Y)] \leq D.$$

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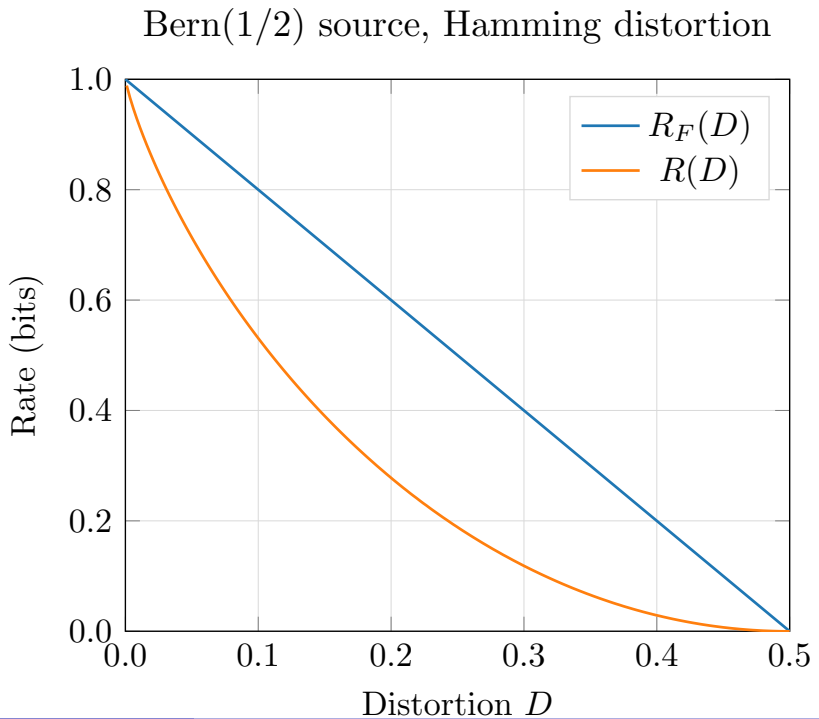
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For any source P_X and $D > 0$,

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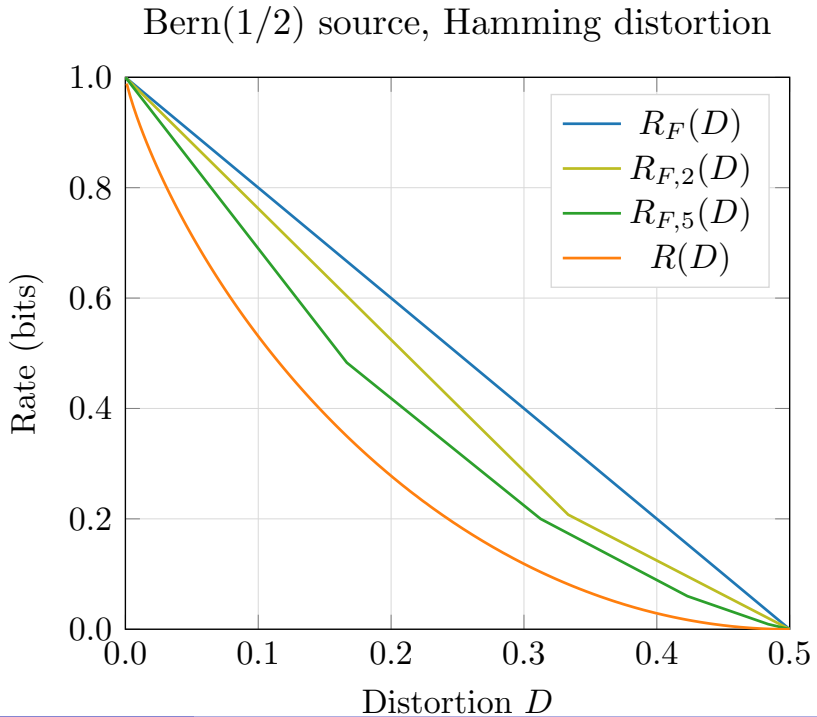
Theorem

For any source P_X and $D > 0$,

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Similarly, with $R_{O,n}(D) := \frac{1}{n} R_O(nD)$ and $R_{F,n}(D) := \frac{1}{n} R_F(nD)$, for any P_{X^n}

$$R_{F,n}(D) \leq R_{O,n}(D) \leq R_{F,n}(D) + \frac{3.45}{n}.$$



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ITW Paper!